

WELL BEGUN IS HALF DONE: A NEW INITIALIZATION STRATEGY FOR TARGETED DECISION-BASED ATTACK: SUPPLEMENTARY MATERIAL

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1. OUTLINE

The supplementary document is organized as follows: Section 2 presents an ablation study on filters (DCT vs Gaussian filter across varying cut-off parameters); Section 3 reports complete results on ImageNet, including comparison with the Copy-pasting initialization strategy [1]; and Section 4 provides results on the CIFAR10 dataset.

2. ABLATION STUDY ON FILTERS

We first make an ablation study on the choice of low-frequency (LF) filters and their parameters. Specifically, we execute TtBA [2] against the ResNet50 model, and let the cut-off frequency γ of the DCT-based filter range from 0.05 to 0.3, and the radius k of the spatial filter (Eq. (4) in the main text) range from 1 to 6. A smaller γ or a larger k results in a more blurry image. Figures 1(a) and (b) depict the relationship of the achieved average L_2 distance (after 10000 queries) and the filter parameter for the DCT-based filter and spatial filter, respectively. Both filters show comparable, robust improvements over vanilla initialization, with optimal parameter ranges of $\gamma \in [0.05, 0.15]$ and $k \in [4, 6]$. In our study, we select the spatial filter and set $k = 4$ for simplicity ($k = 1$ for CIFAR10).

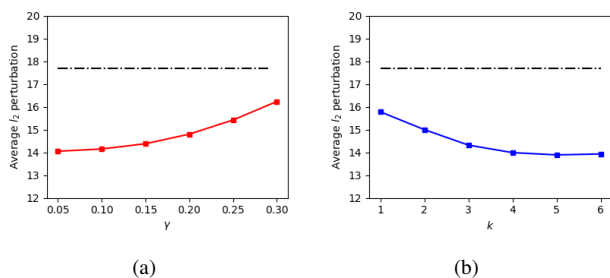


Fig. 1. Effect of the filter parameters on the L_2 distance after 10000 queries. (a) DCT-based filter, (b) Spatial filter. The black line denotes the achieved L_2 distance by the baseline.

3. RESULTS ON IMAGENET

3.1. Comparison with Copy-pasting

In [1], Brunner et al. proposed a copy-pasting strategy to initialize targeted decision-based attacks, which is the only work dedicated to this long-neglected topic to our best knowledge. Unlike our plug-and-play approach, [1] requires **multiple target-class images** (x_{tar-s}), and an **auxiliary surrogate model**. Despite these distinct prerequisites, we are interested in comparing the proposed method with the copy-pasting strategy. Following the setup in Section 4.1 of the main text, we execute CGBA-H [3] on the ImageNet competition dataset using the copy-pasting initialization, equipping the attacker of [1] with GradCAM [4] saliency maps extracted using a pretrained ResNet18 model. To make the comparison as fair as possible, we assure consistent (x_{ori}, x_{tar}) pairs for all competitors (which means the attacker cannot choose what he wants from a large pool of x_{tar-s} as [1]). Table 1 compares the median and average L_2 norm achieved by different initialization strategies across various query budgets. Even without surrogate model guidance, the proposed WellBegun strategy outperforms the Copy-pasting strategy in general. However, the reported results do not diminish the contributions of [1]. Since our LF subspace fusion and their attention-guided pasting address initialization from orthogonal perspectives, integrating both strategies offers a promising direction for future research.

Fig. 2 provides a side-by-side comparison of different initialization strategies for three samples, highlighting their distinct properties. The Copy-pasting strategy moves x_{start} closer to x_{ori} by leaving non-salient regions untouched. Such an advantage is significant when just a small area of x_{tar} is enough to mislead the victim model, as shown in the last row. In contrast, the proposed method provides a closer x_{start} by attenuating most LF components of x_{tar} , delivering consistent, robust gains across all three samples.

3.2. Results on more victims

We report the complete results on ImageNet here, some of which are omitted from the main text due to space limitations. Table 2 reports the median (and average) distortion when attacking three convolutional models and two transformer-

Victims	Attacks	2000 queries	5000 queries	10000 queries
Res50	CGBA-H	51.54(51.85)	33.15(33.80)	17.56(19.44)
	Copy-pasting+CGBA-H	41.07(43.39)	25.75(28.96)	14.48(17.14)
	WB+CGBA-H	27.79(31.98)	19.11(22.50)	12.08(14.62)
Inc-V3	CGBA-H	51.90(55.71)	36.01(40.37)	23.01(27.87)
	Copy-pasting+CGBA-H	47.16(49.12)	36.73(38.83)	27.79(29.92)
	WB+CGBA-H	32.06(37.18)	24.52(28.05)	17.64(20.28)
Res50adv	CGBA-H	59.81(63.11)	47.03(48.96)	32.62(35.22)
	Copy-pasting+CGBA-H	48.88(52.92)	38.80(42.19)	28.82(31.88)
	WB+CGBA-H	40.95(44.99)	31.77(36.17)	25.02(28.08)

Table 1. Median (average) distortion comparison for different initialization schemes. Dominant results are in **bold**.

based models. The transformer-based models seem to be more susceptible to decision-based attacks, consistent with the results reported in [2]. For example, the median distortion after 10000 queries for all attacks are below 10 when attacking transformer-based models. Nevertheless, our method demonstrates its advantage in most cases, especially when the query budget ≤ 5000 .

4. RESULTS ON CIFAR10

Since the proposed initialization scheme relies on the frequency decomposition of input samples, we are wondering whether it is still effective on low-resolution images. To this end, we conduct an experiment on the CIFAR10 validation dataset. Tab. 2 compares the L_2 norm achieved by different methods across query budgets of 1000, 2000, and 3000. The results demonstrate that the proposed strategy remains advantageous on low-resolution images, especially when the query budget is strict. For example, when attacking the ResNet20 model, WB+ reduces the median distortion of different attacks by 18.5% (TtBA [2], $1.35 \rightarrow 1.10$) $\sim 24.9\%$ (HSJA[6], $2.81 \rightarrow 2.11$) within 1000 queries.

5. REFERENCES

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Fig. 2. Visual comparison of the starting points with different strategies. From left to right, x_{ori} , x_{tar} , x_{start} s with the default direct blending strategy, the Copy-pasting strategy [1], and the proposed method.

Victims	Attacks	2000 queries	5000 queries	10000 queries	Attacks	2000 queries	5000 queries	10000 queries
Res50	HSJA	51.01(51.30)	30.05(30.24)	15.55(17.32)	CGBA-H	51.54(51.85)	33.15(33.80)	17.56(19.44)
	WB+HSJA	33.19(37.39)	21.32(25.08)	13.01(15.70)	WB+CGBA-H	27.79(31.98)	19.11(22.50)	12.08(14.62)
	Bounce	47.85(49.58)	27.29(30.07)	15.15(17.19)	TtBA	50.68(50.99)	31.34(32.11)	14.80(17.72)
	WB+Bounce	29.33(34.85)	20.65(24.01)	12.99(15.44)	WB+TtBA	28.72(31.73)	17.30(22.15)	10.82(13.89)
Inc-V3	HSJA	53.86(56.48)	33.06(35.84)	18.77(22.11)	CGBA-H	51.90(55.71)	36.01(40.37)	23.01(27.87)
	WB+HSJA	37.55(41.45)	26.41(28.86)	16.98(19.32)	WB+CGBA-H	32.06(37.18)	24.52(28.05)	17.64(20.28)
	Bounce	48.99(51.58)	32.33(33.90)	20.13(22.29)	TtBA	52.44(54.25)	34.14(38.43)	19.59(24.79)
	WB+Bounce	33.29(36.89)	23.61(26.38)	17.15(18.88)	WB+TtBA	31.22(35.33)	22.35(26.14)	14.21(18.19)
Swin-T	HSJA	28.04(32.63)	12.77(14.80)	5.82(6.98)	CGBA-H	21.64(24.31)	9.10(11.12)	4.20(5.23)
	WB+HSJA	20.40(23.47)	9.80(11.93)	4.86(6.17)	WB+CGBA-H	13.58(16.86)	7.31(8.67)	3.69(4.50)
	Bounce	24.26(28.55)	11.52(13.48)	5.38(6.56)	TtBA	23.57(26.30)	9.62(12.30)	4.08(5.27)
	WB+Bounce	16.57(19.31)	8.20(10.00)	4.30(5.32)	WB+TtBA	14.67(17.55)	7.74(9.25)	3.76(4.81)
ViT	HSJA	34.51(38.53)	16.34(18.71)	9.21(10.76)	CGBA-H	21.69(24.10)	9.17(10.95)	4.83(5.90)
	WB+HSJA	33.89(36.04)	16.73(19.07)	9.68(11.21)	WB+CGBA-H	19.19(21.56)	8.97(10.90)	4.81(5.89)
	Bounce	31.66(33.97)	15.92(17.30)	8.99(10.20)	TtBA	23.33(21.25)	10.12(8.74)	5.45(4.37)
	WB+Bounce	28.57(30.47)	15.86(17.22)	9.32(10.71)	WB+TtBA	18.77(20.40)	8.44(9.63)	4.04(5.31)
Res50adv	HSJA	69.15(70.27)	51.78(53.54)	35.53(38.55)	CGBA-H	59.81(63.11)	47.03(48.96)	32.62(35.22)
	WB+HSJA	51.76(54.10)	42.48(44.04)	32.50(33.84)	WB+CGBA-H	40.95(44.99)	31.77(36.17)	25.02(28.08)
	Bounce	63.79(65.83)	49.16(49.94)	33.27(36.20)	TtBA	61.18(63.73)	43.36(48.32)	30.65(33.81)
	WB+Bounce	44.41(48.83)	36.73(40.05)	30.16(31.93)	WB+TtBA	42.66(47.72)	33.20(37.49)	24.50(28.40)

Table 2. Median (average) distortion for targeted attacks under different query budgets. Dominant results are in **bold**.

Victims	Attacks	1000 queries	2000 queries	3000 queries	Attacks	1000 queries	2000 queries	3000 queries
Res20	HSJA	2.81(3.45)	1.78(2.34)	1.45(1.78)	CGBA-H	1.16(1.92)	0.56(0.93)	0.37(0.57)
	WB+HSJA	2.11(2.48)	1.65(1.93)	1.38(1.62)	WB+CGBA-H	0.92(1.41)	0.51(0.78)	0.35(0.53)
	Bounce	2.80(3.39)	1.88(2.28)	1.50(1.75)	TtBA	1.35(2.20)	0.54(0.98)	0.39(0.59)
	WB+Bounce	2.26(2.61)	1.82(2.02)	1.48(1.65)	WB+TtBA	1.10(1.51)	0.52(0.78)	0.36(0.53)
Vgg13_bn	HSJA	2.76(3.35)	2.03(2.36)	1.69(1.88)	CGBA-H	1.48(2.23)	0.88(1.18)	0.65(0.79)
	WB+HSJA	2.54(2.81)	2.03(2.19)	1.72(1.84)	WB+CGBA-H	1.44(1.73)	0.87(1.03)	0.65(0.74)
	Bounce	3.03(3.55)	2.15(2.47)	1.77(1.94)	TtBA	1.70(2.45)	0.90(1.21)	0.66(0.82)
	WB+Bounce	2.53(2.86)	2.10(2.26)	1.76(1.89)	WB+TtBA	1.50(1.80)	0.86(1.05)	0.65(0.77)

Table 3. Median (average) distortion for targeted attacks on CIFAR10. Dominant results are in **bold**.